

Session – 1

Que. 1 -

- Developer: Builds software and applications by writing code.
- Tester: Finds bugs and tests software to make sure it works well.
- Designer: Creates how the software looks and how users interact with it.
- Data Analyst: Studies data to help the company make good choices.
- AI Engineer: Builds smart systems and computer models that learn.

Que. 2 –

Product Companies (They make and sell their own software)

- Google: Makes a search engine to find information on the internet.
- Zoho: Makes business tools to help companies manage their office work.
- Microsoft: Makes the Windows operating system for computers.

Service Companies (They do software work for other companies)

- TCS: Hires out engineers to build custom software for other businesses.
- Infosys: Helps other companies manage and fix their computer systems.
- Accenture: Gives expert tech advice and manages IT projects for other clients.

Que. 3 –

- GitHub
 - Full Form: GitHub
- Slack
 - Full Form: Searchable Log of All Conversation and Knowledge.
- Jira
 - Full Form: Gojira
- IDE
 - Full Form: Integrated Development Environment.
- API

- Full Form: Application Programming Interface.

Que. 4 –

- Python: Used to build smart systems, robots, and analyze data.
- JavaScript: Used to make web pages interactive, like adding buttons you can click.
- Java: Used to make big business programs and Android mobile apps.
- HTML/CSS: Used to build the basic look, layout, and style of websites.
- SQL: Used to store, find, and manage information in databases.

Que. 5 –

- A developer writes code to build and create the software application.
- A tester checks the software to find mistakes and make sure it works perfectly.

Session – 2

Que. 1 –

- Presentation Layer: The user-facing front end that displays information and captures user input.
- Logic Layer: The core "brain" of the application that processes rules, calculations, and data workflows.
- Data Access Layer: The backend infrastructure that handles data storage, retrieval, and communication with the database.

Que. 2 –

- Client: The device or app that asks for information.
- Server: A powerful computer that stores data and sends it back when asked.

Real-Life Example:

Think of ordering food at a restaurant. You ask the waiter for a burger, and the kitchen makes the burger and sends it back to your table.

Que. 3 –

Desktop Apps

1. You must download and install them directly onto your computer (PC or laptop).
2. They can often work offline without any internet connection.

Web Apps

1. You do not need to install them; you just open them using a web browser (like Google Chrome).
2. They always need an active internet connection to work.

Mobile Apps

1. You download and install them on portable devices like smartphones and tablets.
2. They are specially designed for small touchscreens and can easily use your phone's camera or GPS.

Que. 4 -

- MySQL: This is a SQL database.
- MongoDB: This is a NoSQL database.
- PostgreSQL: This is a SQL database.

Session – 3

Que. 1 –

- HTTP
 - Full Form: Hypertext Transfer Protocol.
 - Use: A set of rules to load web pages on the internet.
- HTTPS
 - Full Form: Hypertext Transfer Protocol Secure.
 - Use: A safe version of HTTP that hides your data so hackers cannot see it.
- FTP
 - Full Form: File Transfer Protocol.

- Use: A set of rules used to send files from one computer to another over the internet.
- SMTP
 - Full Form: Simple Mail Transfer Protocol.
 - Use: A set of rules used to send emails between servers.

Que. 2 –

- IPv4 uses short, 32-bit numbers (like 192.168.1.1) and has a limited number of addresses.
- IPv6 uses much longer, 128-bit letters and numbers (like 2001:db8::1) to offer nearly infinite addresses.

Que. 3 –

- IPv4 Address: 142.250.190.46
- IPv6 Address: 2607:f8b0:4005:805::200e

Que. 4 –

- 200: Success.
- 301: Moved Permanently.
- 403: Forbidden.
- 404: Not Found.
- 500: Internal Server Error.

Que. 5 –

- Root Level (.): The invisible dot at the end of the web address that starts the search.
- TLD (com): The last part of the name that groups similar websites.
- Domain (google): The main name chosen by the company to identify itself.
- Subdomain (www): A smaller section created under the main domain to point to a specific web page.

Session – 4

Que. 1 –

- Requirement: Gathering and listing what the software needs to do.
- Design: Planning how the software will look and how its system will be built.

- Development: Writing the actual computer code to create the software.
- Testing: Checking the software for errors and bugs to ensure it works correctly.
- Deployment: Releasing the finished software so users can start using it.
- Maintenance: Fixing new problems and updating the software over time.

Que. 2 –

- Process: Agile divides work into small parts and accepts changes easily; Waterfall follows a strict, step-by-step plan where changes are hard to make.
- Testing: Agile tests the software continuously during the project; Waterfall tests the software only at the end.
- Delivery: Agile delivers working software frequently in small pieces; Waterfall delivers the final product all at once at the very end.

Que. 3 –

- Product Owner (PO): Decides what features the software needs based on what the customer wants.
- Scrum Master: Helps the team work together smoothly and removes any problems that slow them down.
- Development Team (Dev Team): Writes the code, tests the software, and builds the actual product.

Que. 4 –

- A Sprint is a short, fixed period of time during which a Scrum team works to complete a specific set of tasks.

Que. 5 –

- Backlog: A list of all the tasks that need to be done in the future.
- To Do: Tasks chosen for the current work period that have not started yet.
- In Progress: Tasks that the team is actively working on right now.
- Done: Tasks that are completely finished and approved.

Session – 5

Que. 1 -

- Initialize a repo: git init
- Stage all files: git add .

- Commit with a message: `git commit -m "your message"`
- Push to remote: `git push`
- Pull latest changes: `git pull`

Que. 2 –

- `git fetch` downloads the latest changes from the remote server but does not change your local working files.
- `git pull` downloads the latest changes and automatically merges them into your local files right away.

Que. 3 –

- Go to github.com and click New to create a repository named my-first-repo.
- Open your terminal and run these commands.

Que. 4 –

- A Fork is a personal copy of someone else's repository that you create on your own account to freely make changes.
- A Pull Request is a polite request asking the original owner to merge the changes from your copy back into their project.

Que. 5 –

- Create and switch to the new branch
- Make a change and commit it
- Switch back to the main branch
- Merge the new branch into main

